INCH-POUND

MIL-W-80235C
13 September 1991
SUPERSEDING
MIL-W-80235B
2 December 1985

MILITARY SPECIFICATION

WELDING MACHINES, ELECTRICAL RESISTANCE, SPOT, PRESS TYPE, AIR OPERATED, SINGLE PHASE, ALTERNATING CURRENT (AC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single phase, alternating current, air operated, press type, electrical resistance, spot welding machines with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following styles and sizes. The size is determined by the transformer kilovolt ampere (KVA) rating at 50 percent duty cycle. The style and size to be furnished shall be as specified (see 6.2).

Style A - 18 inch throat depth.
Style B - 24 inch throat depth.
Style C - 30 inch throat depth.
Style D - 36 inch throat depth.

Size 1 - 50 KVA.
Size 2 - 75 KVA.
Size 3 - 100 KVA.
Size 4 - 150 KVA.
Size 5 - 250 KVA.

1.2.1 Throat depth. The welding machine throat depth is the clear unobstructed distance, measured in the welding plane, from the centerline of the welding electrodes to the first obstruction of the machine frame.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-12332 - Welding, Resistance, Spot, Seam, and Projection for; Fabricating Assemblies of Low-carbon Steel.

MIL-W-45210 - Welding, Resistance: Spot, Weldable Aluminum Alloys.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards. (DOD adopted)

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete welding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The welding machine shall be new and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

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3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the welding machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2).

3.2.5 Environmental protection. During the manufacture, service transportation, or storage of the machine and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operators health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.6 Lubrication. Means shall be provided to ensure lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion or failure.

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3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the welding machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The welding machine shall consist of not less than the following components.

3.4.1 Machine frame and base. The frame and base assembly of the welding machine shall be a steel weldment of box construction, ribbed to maintain original alignment and to minimize deflection. The base shall provide for floor mounting. Access to the interior shall be provided to permit inspection and servicing of the equipment installed within the frame. Doors and access panels shall be provided with locks and interlocks to prevent access by unauthorized personnel to live portions of the machine. The frame shall be designed to permit positioning of the lower knee and horn assembly.

3.4.2 Welding transformer. The welding transformer shall be mounted in the frame of the machine. The transformer KVA rating and secondary short circuit current output shall be in accordance with table I for the machine size specified. The transformer shall be water cooled and thermostatically controlled. A temperature responsive device shall be provided to prevent power being applied to the transformer if the temperature of the transformer exceeds design limits. All static-connecting surfaces of the secondary circuit shall be silver plated. The transformer insulation shall meet or exceed the requirements of RWMA Bulletin 16, class B.

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TABLE I. Welding machine sizes and capacities.

Machine sizes	Welding transformer power rating (KVA)	Spot welds per minute	Spot weld throat depth (inches)	Electrode force pounds 1/	Secondary short circuit current + 10% (amperes)	Weld capacity two equal thicknesses of SAE 1010 steel (inches)
1	50	11	18	2250	22,000	0.109
		15	24	1550	19,000	0.094
		16	30	1550	17,000	0.094
		18	36	1000	19,000 2/	0.074
2	75	8	18	2250	27,000	0.141
		18	24	1550	23,000	0.104
		23	30	1000	23,000 2/	0.091
		28	36	1000	23,000 2/	0.074
3	100	5	18	4000	31,000	0.220
		7	24	3000	27,000	0.187
		11	30	2250	27,000 2/	0.164
		15	36	2250	27,000 2/	0.141
4	150	3	18	5000	38,000	0.312
		3	24	5000	34,000	0.312
		2	30	5000	34,000 2/	0.312
		2	36	5000	34,000 2/	0.312
5	250	9	18	5000	50,000	0.312
		5	24	5000	45,000	0.312
		3	30	5000	45,000 2/	0.312
		3	36	5000	45,000 2/	0.312

1/ Based upon air pressure of 80 psi.

2/ These values are based upon the use of transformers having higher open circuit secondary voltages than transformers used with machines having shorter throat depths, with the same KVA rating.

3.4.3 Controls. Operator controls shall be located so as to be readily accessible to the operator. All calibrations, phase shift controls, and similar characteristics shall be clearly marked.

3.4.3.1 Operator control panel. The operator control panel shall have an on-off switch, a power "on" and sequence lights, a current adjustment control, a weld timer selector, auxiliary and accessory control function switches, and any other controls or switches required to meet the requirements specified herein.

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3.4.3.2 Welding power control. The machine shall be furnished with a solid state synchronous precision combination welding power control unit. The control unit may be located in a separate cabinet, or mounted as an integral part of the machine. Unless otherwise specified (see 6.2), a pulsation option shall be included on machines welding material thickness of 1/8 inch and above.

3.4.3.3 Timing and sequencing controls. The control equipment shall provide operating control and adjustment of the welding variables to ensure that optimum values of force and current timing and sequencing can be repetitively attained as necessary to produce spot welds in accordance with this specification. Unless otherwise specified (see 6.2), the machine shall be furnished with a sequence timer. The timing and sequencing functions shall be selected and adjusted by means of knobs, push buttons or digital switches, mounted on the front or side of the machine, so as to be readily accessible and in full view of the operator. The timing control input shall be calibrated in cycles, based on the electrical service line frequency of 60 Hz. The timer shall have at least four functions: "squeeze", "weld", "hold", and "off". When specified (see 6.2), the controls shall be equipped with functions to perform welds specified in MIL-W-45210 for aluminum alloys. Additional control functions shall be furnished as specified (see 6.2). The controls shall have signal lights on the panel to indicate the operating sequence of the welding cycle.

3.4.3.4 Current adjustment control. Unless otherwise specified (see 6.2), continuously variable adjustment of the welding current, from the upper to the lower limits of the machine capacity, shall be provided. The control shall be graduated from 0 to not less than 99 or from 20 percent to not less than 99 percent for convenience in identifying settings. The means of current adjustment provided shall be such that, at the maximum setting of the input, the machine maximum short-circuit current produced is not less than the value specified in table I.

3.4.3.5 Foot switch. A foot switch which initiates the welding cycle shall be furnished. When specified (see 6.2), a two stage switch shall be supplied which initiates the electrode force in the first stage, and the welding sequence in the second stage. The foot switch shall be guarded to prevent accidental operation of the machine.

3.4.3.6 Tip dress control. A tip dress control shall be furnished. This control shall allow the upper electrode to be lowered under a reduced force for cleaning and aligning the electrodes.

3.4.3.7 Weld/no weld switch. A weld/no weld switch shall be furnished. In the no weld position, the control sequence will actuate except that no welding current shall flow.

3.4.3.8 Repeat on/off (repeat/non-repeat) switch. A repeat on/off switch shall be furnished. In the off or non-repeat position only one welding cycle will result with the actuation of the switch. In the on or repeat position, the welding cycle will repeat automatically after a controlled time off interval as long as the switch remains closed.

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3.4.3.9 Electrode forge force selector. When specified (see 6.2), the machine shall have an electrode forge force selector that permits selection of a constant high, constant low, and a variable force sequence.

3.4.4 Welding contactor. Unless otherwise specified (see 6.2), the welding contactor shall be a silicon controlled rectifier (SCR) type. The SCR contactor shall be mounted in a heat sink and shall be water cooled and thermostatically protected. Means shall be provided to prevent welding if the water is shut off or if the supply is restricted. The SCR contactor shall be protected from excessive current demands. The firing circuit element shall be protected by a fuse or other protective device, and a light shall be provided to indicate when the firing circuit has been opened by the protective device.

3.4.5 Automatic voltage compensator. Unless otherwise specified (see 6.2), the machine shall have an automatic voltage compensator (AVC) system to minimize the effects of line voltage variations. The AVC shall maintain the welding current within ± 2 percent, for a line voltage variation of ± 10 percent. Based on the input line frequency, full compensation shall occur within 1-1/2 Hz for a sudden change in line voltage.

3.4.6 Pneumatic system. The pneumatic system and components shall conform to the applicable requirements of RWMA Bulletin Number 16.

3.4.6.1 Pneumatic system operation. Unless otherwise specified (see 6.2), the pneumatic system air cylinder shall be a double action adjustable stroke type. The inlet connection shall be marked for the pressure requirements. The ram and upper electrode pneumatic system shall include valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring switches, and all related equipment necessary to permit use of any of the electrode forces. Air cushions or air speed valves shall be provided to insure minimum mechanical or pneumatic shock operation. The system shall be controlled by a solenoid valve actuated by a foot switch. A means for lowering the upper electrode under reduced force for cleaning and aligning the electrodes shall be provided.

3.4.6.2 Air pressure indicating devices. Unless otherwise specified, the air pressure indicating devices shall be calibrated in pounds per square inch gage (psig). When specified (see 6.2), calibration shall be in pounds of applied electrode force. Calibration accuracy shall be not less than ± 2 percent of full scale. The indicating device shall be mounted in a location which is visible to the operator.

3.4.6.3 Air regulators. The air regulators shall be the self-bleeding type. The regulators shall maintain the pressure at ± 1/2 psi, as long as the supply pressure is not less than 2 psi higher than the preset regulator pressure. The air system for the upper electrode shall produce the required electrode force to perform the welds specified herein. When a variable electrode forge force selector is furnished (see 3.4.3.9), the rise of the electrode force, from its weld force level to its forge force level, shall be 75 percent accomplished in no more than 10 milliseconds

3.4.7 Lower knee and horn. The lower knee and horn shall be mounted on the frame of the machine to provide a vertical adjustment range of not less than 6 inches. A hydraulic jack shall be permanently mounted on the base of

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the machine to aid in the adjustment of the arm. The knee shall have a socket to permit motion of the horn and alignment to the lower electrode holder. The socket shall have a clamp that will not mar the horn. The horn shall be made of solid copper and have a diameter as specified in table II. The horn shall have clamps that will not mar the electrode holders. With maximum rated electrode force applied, and with the upper horn and lower horn unsupported by any bracing means, the difference between the horizontal deflection of the upper electrode and the lower electrode shall not exceed the limits specified in table III. The vertical deflection of the lower electrode holder shall not be more than the value specified in table III.

TABLE II. Welding machine sizes and dimensions.

Size	Welding transformer power rating (KVA)	Standard ram stroke including adjustments	Horn diameter	Electrode holder diameter	Electrode holder/ taper #
1	50	3	2.25 to 2.75	1.25	RWMA 5/ # 2 Morse
2	75	3	2.25 to 2.75	1.25	RWMA 5/ # 2 Morse
3	100	3	2.75 to 3.00	1.50	RWMA 7 # 3 Morse
4	150	4	3.00 to 3.50	1.50	RWMA 7 # 3 Morse
5	250	4	3.00 to 3.50	1.50	RWMA 7 # 3 Morse

NOTE: All dimensions are in inches.

TABLE III. Maximum permissible electrode deflection.

Machine sizes	Horizontal deflection deflection	Vertical deflection of lower electrode holder			
	All throat depths	Throat depth			
		18	24	30	36
1 and 2	0.005	0.011	0.016	0.021	0.021
3	0.007	0.015	0.020	0.025	0.025
4 and 5	0.010	0.021	0.026	0.031	0.031

NOTE: All dimensions are in inches.

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3.4.8 Head and ram assembly. The head and ram assembly shall be designed and constructed to maintain alignment and conform to the deflection requirements specified in table III. The guiding mechanism for the ram shall be of the antifriction type, provided with adjustment for wear, and shall be protected against dust and foreign matter. The ram movement shall be operated by an air cylinder that is actuated by a foot switch. The upper electrode shall be electrically connected to the secondary windings of the transformer by a silver plated, laminated flexible shunt. The ram stroke (retraction) shall conform to the requirements specified in table II. The ram shall be cushioned for minimum shock operation.

3.4.9 Electrodes and holders. The electrodes and holders shall conform to table II and to RWMA Bulletin 16. Unless otherwise specified (see 6.2), one set of electrode holders 8 inches in length and one set of straight electrodes of RWMA group A, class 2 alloy with a style F nose shall be supplied.

3.4.10 Water cooling system. The machine shall have a water cooling system that will prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply and shall have a single inlet connection into a manifold. The inlet connection shall be marked for pressure and flow requirements. The system shall have parallel flows from the manifold to the welding transformer, to the secondary winding and circuit conductors, to the primary welding current contactors, and to the upper and lower electrodes. The piping in the system shall be corrosion resistant to tap water service. The system shall have an inlet strainer and a closed manifold drain system with flow indicators. The water flow lines to the electrodes shall have solenoid valves to prevent water flow while changing electrodes. When specified (see 6.2), the system shall have water saving devices.

3.4.11 Electrical requirements. The machine shall be wired to conform to NFPA-70. Unless otherwise specified (see 6.2), the machine shall operate on single phase, 60 Hz, 230 or 460 volts, primary electric service. The welding machine shall be initially wired for operation on 460 volts.

3.4.12 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated time of welding current flow. The meter shall be of the non-resetting type and shall have a range of not less than 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of tables I, II, and III for the size specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

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3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance requirements. The welding machine shall conform to all requirements of this specification and shall produce spot welds conforming to the requirements of MIL-W-12332. When specified (see 6.2), the machine shall meet the requirements of MIL-W-45210.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.10.1 Lubrication plate. When specified (see 6.2) a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2) the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below and that information required in RWMA Bulletin Number 16. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)

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Amp rating of largest motor
Short circuit/over current protection rating
Contract number or order number
National stock number
Date of manufacture

3.11 Workmanship. Workmanship of the welding machine and accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.7.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6, and all tests in 4.7. The welding machine shall pass the examination and all tests to be accepted.

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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1 , and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. Each welding machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. Thirty consecutive welds shall be made using two thicknesses of the material specified in table I for the machine size furnished. The welds shall be examined and tested in accordance with MIL-W-12332. When aluminum alloys welding capacity is specified, in accordance with 3.6, 30 consecutive welds shall be made and tested in accordance with MIL-W-45210. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.3 Electrode deflection test. The machine shall be tested to determine the vertical and horizontal deflection of the upper and lower electrodes. The electrodes shall be positioned at the nominal throat depth location and the pneumatic system adjusted to provide the machine maximum electrode force value specified in table I. Without the lower horn being supported by a brace or any other device that can serve as a brace, the electrodes shall be brought to bear upon a plate and roller assembly as illustrated in figure 1. The diameter of the rollers and the thickness of the plates shall be not less than one-half the electrode diameter and preferably not more than the diameter. Under these conditions, the difference between the upper electrode and the lower electrode horizontal deflection and the vertical deflection of the lower electrode holder shall be not more than specified in table III.

4.7.4 Automatic voltage compensator test. When the AVC is furnished (see 3.4.5), the performance of the AVC shall be measured by means of a calibrated recording instrument. With the current adjustment set first at 50 percent and then at 90 percent of maximum, the secondary current and primary voltage shall be recorded as ± 10 percent step changes are introduced into the primary voltage. Alternately, the compensation and response time may be tested by causing the voltage change to occur only on the control of the AVC, in which case, the 10 percent increase or decrease in voltage cause, respectively, a decrease or increase of 10 percent in the secondary current within ± 2 percent. The secondary current compensation and response time shall be as specified in 3.4.5.

4.7.5 Forge force rise time. When the electrode force control is furnished, the machine sequence shall be set in the variable force mode, and with the machine controls adjusted to the proper values for welding two pieces of the maximum thickness of material specified in table I, the rise time for

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the forge force and the time relationship of the weld current and the forge force shall be measured by means of a calibrated recording instrument. The incremental rise of the electrode force in the variable pressure mode from its weld force level to its forge force level shall be 75 percent accomplished in no more than the time specified in 3.4.4.3.

4.7.6 Secondary current test. The secondary maximum current shall be determined in accordance with RWMS Bulletin Number 16. The output current shall comply with values specified in table I.

4.7.7 Rated capacity test. The welding machine shall be adjusted for welding the maximum thickness of material specified in table I to produce its rated current at 50 percent duty cycle and operated for 1 hour of simulated welds. The temperature limits specified in RWMA Bulletin Number 16 shall not be exceeded during the test. There shall be no evidence of overheating, arcing or malfunction of the welding machine.

4.7.8 Dielectric test. The machine shall be subjected to the dielectric test described in RWMA Bulletin Number 16.

4.7.9 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The welding machines covered by this specification are intended for spot welding.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

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e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.4).

g. Environmental protection requirements exceeding EPA/OSHA requirements, if required (see 3.2.5)

h. Painting if different (see 3.3.5.).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

j. Pulsation option, if different (see 3.4.3.3).

k. Sequence timer, if different (see 3.4.3.3).

l. Control functions for welding aluminum alloys, if required (see 3.4.3.3).

m. Additional timing and sequencing control functions, (see 3.4.3.3).

n. Current adjustment control, if different (see 3.4.3.4).

o. Two stage foot switch, if required (see 3.4.3.5)

p. Electrode forge force selector, if required (see 3.4.3.9).

q. Welding contactor, if different (see 3.4.4).

r. Automatic voltage compensator, if different (see 3.4.5).

s. Pneumatic system air cylinder, if different (see 3.4.6.1).

t. Pressure indicating device, if required calibrate in pounds of applied electrode force (see 3.4.6.2).

u. Electrodes and holders, if different (see 3.4.9).

v. Water saving devices, if required (see 3.4.10).

w. Electric requirements, if different (see 3.4.11).

x. Hour meter, if required (see 3.4.12).

y. Size and capacity, if different (see 3.5).

z. Physical size limitations, if required (see 3.5.1).

aa. Conform to MIL-W-45210, if required (see 3.6).

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bb. Additional performance features, if required (see 3.6.1).

cc. Specify and describe optional equipment, if required (see 3.8).

dd. Specify applicable repair parts, if required (see 3.9).

ee. Lubrication plate, if required (see 3.10.1).

ff. Lubrication plate, if different (see 3.10.1).

gg. Nameplate, if different (see 3.10.2).

hh. First article inspection, if different (see 4.4).

ii. Quality conformance inspection, if different (4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and

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disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

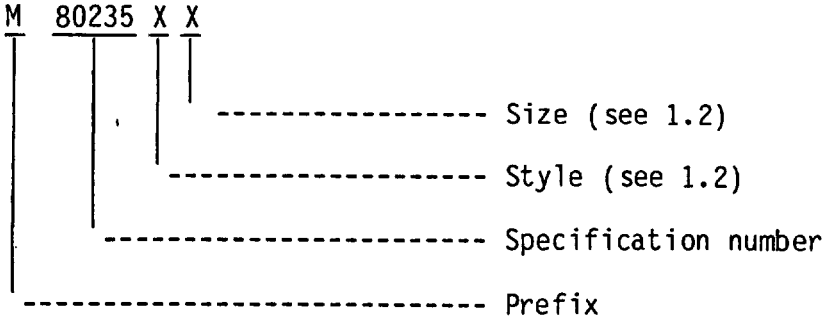
6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Part or Identifying Number (PIN). The PIN to be used for welding machines acquired to this specification are created as follows:



6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

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6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Subject term (key word) listing.

Bonding
Electric welding
Electric welding machines
Joining
Pressure welding
Metal bonding
Metal joining
Spot welding machine
Weld metal

Custodian:
Air Force 99

Review activities:
Air Force - 84
DLA - GS

User actvities:
Navy - OH, AS, CG

Preparing activity:
DLA-IP

Project 3432-0144

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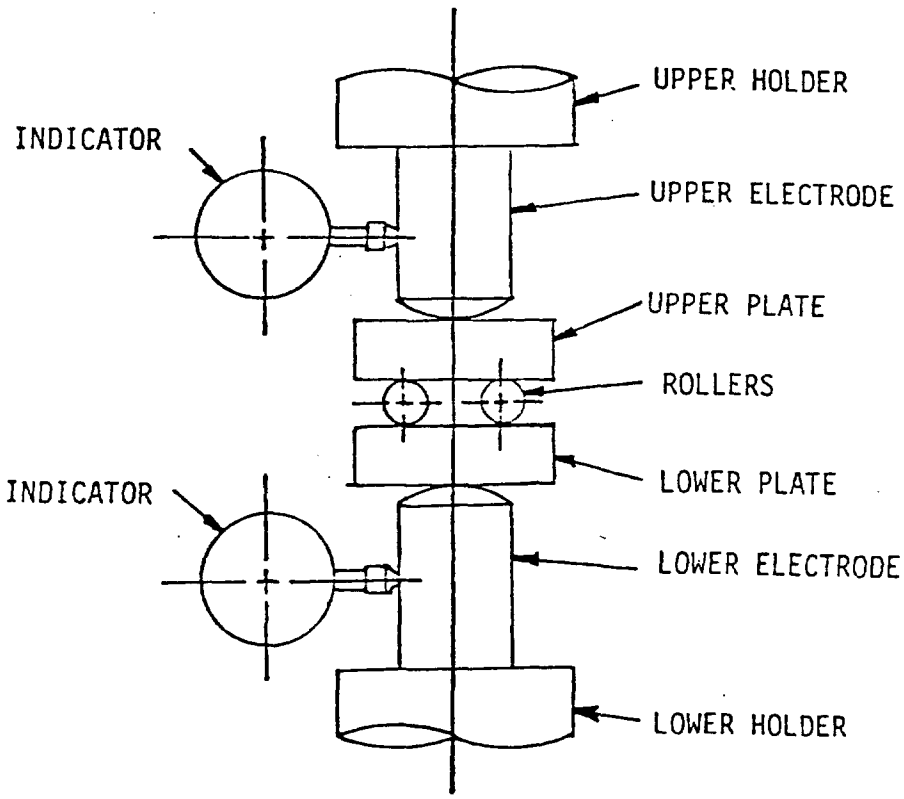


FIGURE 1. DETAIL OF DIAL INDICATOR SET-UP FOR MEASURING ELECTRODES HORIZONTAL DEFLECTION

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-80235C	2. DOCUMENT DATE (YYMMDD) 91 09 13

3. DOCUMENT TITLE WELDING MACHINE, ELECTRICAL RESISTANCE, SPOT, PRESS TYPE, AIR OPERATED, SINGLE PHASE, ALTERNATING CURRENT (AC)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	b. TELEPHONE (Include Area Code) (1) Commercial: (901)-775-4749 (2) AUTOVON: 683-4749
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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